Self-Supervised Random Forest on Transformed Distribution for Anomaly Detection

Jiabin Liu, Huadong Wang, Hanyuan Hang, Shumin Ma, Xin Shen, and Yong Shi, *Senior Member, IEEE*

***Abstract*— Anomaly detection, the task of differentiating abnormal data points from normal ones, presents a significant challenge in the realm of machine learning. Numerous strategies have been proposed to tackle this task, with classification-based methods, specifically those utilizing a self-supervised approach via random affine transformations (RATs), demonstrating remarkable performance on both image and non-image data. However, these methods encounter a notable bottleneck, the overlap of constructed labeled datasets across categories, which hampers the subsequent classifiers' ability to detect anomalies. Consequently, the creation of an effective data distribution becomes the pivotal factor for success. In this article, we introduce a model called "self-supervised forest (sForest)," which leverages the random Fourier transform (RFT) and random orthogonal rotations to craft a controlled data distribution. Our model utilizes the RFT to map input data into a new feature space. With this transformed data, we create a self-labeled training dataset using random orthogonal rotations. We theoretically prove that the data distribution formulated by our methodology is more stable compared to one derived from RATs. We then use the self-labeled dataset in a random forest (RF) classifier to distinguish between normal and anomalous data points. Comprehensive experiments conducted on both real and artificial datasets illustrate that sForest outperforms other anomaly detection methods, including distance-based, kernel-based, forest-based, and network-based benchmarks.**

***Index Terms*— Anomaly detection, data distribution, random forest (RF) classifier, random Fourier transform (RFT), random orthogonal rotations, self-supervised learning.**

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I. Introduction

DEEP learning, with its demonstrated efficacy in a plethora of domains, heavily depends on the availability of large-scale, high-quality, and accurately labeled data to effectively train deep neural networks. However, due to privacy and security issues, it is usually difficult to obtain trustworthy ground-truth labels in practice [1], [2]. For instance, in the field of cybersecurity, identification of novel, unknown attacks is frequently hampered due to the absence of patterns or signatures of these attacks, which are essential for a supervised classifier. Similar problems arise in industrial detection, wherein we often find ourselves unable to anticipate all possible fault situations. This shortage of labeled data considerably impedes the process of supervised learning.

Anomaly detection, also referred to as outlier detection, fault detection, or novelty detection, emerges as a solution to the challenges previously discussed. The key concept involves training exclusively with "normal" samples and identifying "abnormal" samples within the test dataset. This is accomplished by labeling all test samples as either normal or anomalous. Researchers have successfully designed effective anomaly detection techniques applicable across a myriad of fields. For instance, within the realm of computer vision, the techniques have been applied to identify abnormal trajectories and moving objects [3], [4], [5], [6]. Within the finance sector, anomaly detection plays a crucial role in fraud detection [7], [8], [9], [10], [11], [12]. Moreover, in healthcare and medicine, the technique proves useful for monitoring and addressing abnormalities within medical records [13]. For a more comprehensive understanding of this topic, readers are encouraged to refer to the surveys in [14] and [15].

Self-supervised learning has recently emerged as a promising approach, demonstrating an impressive capability to generate better feature representations. Techniques widely employed in self-supervised learning for image data encompass geometric transformations [16], [17], [18], patching [19], [20], and color permutations [21], [22]. This learning paradigm has recently been adopted to address anomaly detection problems [17], [23] for image data. Here, image rotations are leveraged to generate distinct categories, and a neural network is trained based on these "supervised" data. Since these pretext tasks are largely inapplicable to the analysis of structured data, Bergman and Hoshen [24] broadened the application of classification-based pretext tasks to outlier detection with general data by defining multiple random affine transformations (RATs) as different categories. It is noteworthy, however, that

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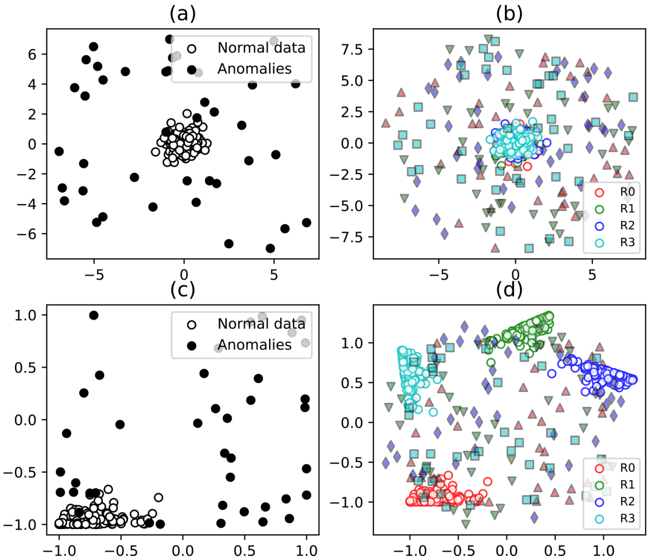

Fig. 1. Effect of data distribution in a 2-D space. (a) Original data. (b) Rotated data based on the original data in (a), where the transformed abnormal data is specially marked by adding a black edge. (c) Transformed data based on a random Fourier transform (RFT) $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. (d) Rotated data based on the transformed data in (c). The $\{R_i\}_{i=0}^3$ represent the four categories of self-labeled data, and each category corresponds to a random rotation transformation.

the distribution of data procured from affine transformation is of great importance, as it directly influences the performance of the final classifier. As an illustration, we display the effect of kernel feature transformation in a 2-D space in Fig. 1. In Fig. 1(b), the four categories of self-supervised data are intensely mixed, making it challenging for the subsequent classifier to distinguish features and thereby leading to failure in anomaly detection. Therefore, we propose that an optimal approach to implementing self-supervised-based anomaly detection on general data involves "creating" a superior data distribution for the ensuing classifier.

In this article, we introduce a novel distribution for self-supervised-based anomaly detection, which incorporates RFT [25] and random orthogonal rotations. This new methodology supplants the RAT, previously used in research [24], thereby enabling more effective control over the distribution of the generated data. Our proposed method initially maps the input data into a feature space at the onset of the learning process, applying RFT [25] to the original data. Subsequently, these newly generated, less correlated, or redundant features are used to train a random forest (RF) classifier via supervised learning with self-labeled data. This self-labeled data is produced through random orthogonal rotations. In this article, we primarily employ our data distribution methodology with RF to underscore the advantages of our data distribution. However, it is essential to note that our data distribution can be applied to any classifiers, including neural networks, not just the RF classifier. In Fig. 1(d), there is a pronounced separation between the four categories of normal data following transformation. Conversely, the transformed abnormal data points intermingle. During the testing phase, the same series of rotations will be implemented on the testing data, which includes both normal and anomalous samples. The trained RF classifier will then be used on the self-labeled testing data. Ultimately, samples with low testing accuracy are categorized as anomalies.

In summary, the main contributions of this article are mentioned as follows.

1) We develop a self-supervised anomaly detection method for structured data, dubbed "self-supervised forest (sForest)." This method introduces a novel self-labeled data distribution, derived from the application of RFT and orthogonal rotation, for self-supervised anomaly detection methodologies.
2) We provide a theoretical rationale for our method's superiority. The data distribution obtained through the combination of RFT and orthogonal rotation is more stable ("kernel approximation with lower expected variance") than that acquired through RAT.
3) The superiority and robustness of our proposed method have been examined on various artificial and benchmark datasets. The performance of "sForest" is contrasted with several state-of-the-art anomaly detection algorithms, encompassing forest-based methods isolation forest (iForest) and partial identification forest (PIDforest), distance-based methods [i.e., k-nearest neighbor (k-NN) and local outlier factor (LOF)], kernel-based method one-class support vector machine (OCSVM), and the self-supervised-based method geometric-transformation anomaly detection (GOAD).

II. Related Works

The study of anomaly detection, alternatively termed as outlier detection, fault detection, or novelty detection, has gained substantial traction within the field of machine learning. Broadly, it encompasses three distinct methodologies: supervised, unsupervised, and semi-supervised learning, the choice of which hinges on the accessibility of labeled data. In this research paper, our primary focus rests on semi-supervised anomaly detection, a scenario wherein only a minimal number of samples are labeled, leaving the majority without ground-truth labels. Notably, based on the strategies applied to this learning paradigm, it can be segmented into distance-based methods [26], [27], [28], density estimation and probabilistic models [29], [30], as well as classification-based methods [17], [23], [24], [31].

In recent years, the field of anomaly detection has witnessed an evolutionary leap, wherein the self-supervised mechanism within this learning paradigm has been leveraged to reframe anomaly detection as a classification problem. This novel approach has delivered substantial enhancements in performance across both image and non-image datasets. In fact, it transforms the structured data into different categories by defining a set of geometric transformations and makes the anomaly detection problem a conventional classification task.

For the image data, Golan and El-Yaniv [17] initially introduced the concept of a "supervised task" by rotating each image by 90°, 180°, and 270°, where each angle is defined as an individual category. This approach was later expanded upon by Hendryck and Gimpel [23], who combined additional self-supervision mechanisms to enhance performance. They deemed the low-confidence samples from the softmax activation statistics to be anomalies, following a specific selection rule. However, these geometric transformations are primarily designed for unstructured data like images and are therefore not directly transferable to tabular data.

Addressing the challenge of non-image data, Bergman and Hoshen [24] broadened the scope of the geometric transformation method to include RATs. They put forth an anomaly detection technique that synergistically amalgamated a one-class model with the transformation classification method. This methodology boasts a dual advantage: it can process a wide array of data types and has an auto-tuning capability for transformation parameters, thanks to the utilization of RATs. Furthermore, the variance between disparate data types can be effectively minimized through the selection of random transformations, simultaneously reducing the detrimental impact of adversarial samples.

However, the method proposed by [24] has overlooked the influence of data distribution on the overall performance of the self-supervised model. In terms of the geometric morphology of the original data distribution, the application of affine transformation functions preserves a similar morphology in the transformed data. Moreover, if the geometric morphology of the original data distribution is unsuitable, the self-labeled data after transformation will be challenging for classifiers to discern (as shown in Fig. 1, where the original data is normally distributed around the origin), or substantial intervals between different classes of data are generated. This allows classifiers to easily differentiate between data classes, which complicates the learning of complex patterns of normal data distribution for any classifier, leading to the failure of anomaly detection. In addition, this method does not consider the influence of data redundancy or feature correlation on the self-supervised model.

III. Algorithm Design

We train the sForest model only on the normal data $\mathcal{D} := \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, where each observation $\mathbf{x}_i$ is drawn i.i.d. from one probability distribution P. The construction of sForest starts with mapping the input data into a feature space $\mathcal{Z} := \{\psi(\mathbf{x}_1), \dots, \psi(\mathbf{x}_N)\}$, with a nonlinear kernel transformation $\psi(\cdot)$ (such as the Monte Carlo approximation of the Fourier transform [25]). Then, we augment the data by applying random rotations $\{R_m\}_{m=0}^{M}$ to the features. In particular, R_0 signifies the identity transformation, implying that no changes are made to the features when the rotation R_0 is applied. For every rotated feature, $R_m(\psi(\mathbf{x}_i))$, is assigned the corresponding label R_m, resulting in a self-labeled dataset: $\{(R_m(\psi(\mathbf{x}_i)), R_m)\}_{i\in\{1,\dots,N\}, m\in\{0,\dots,M\}}$. Using these self-labeled data pairs, we proceed to train an RF classifier in a supervised way. During the testing phase, similar operations are carried out on the test data using the function $\psi(\cdot)$ and the set of rotation matrices $\{R_m\}_{m=0}^{M}$. The previously trained RF classifier is utilized to assess the samples. Anomalies are discerned based on low testing accuracy.

A. Rotations and RFT in Self-Supervised Learning

1) Rotations in Self-Supervised Learning: For the image data, classical self-supervised learning algorithms usually perform fixed-angle rotations with the angle θ taking values from the set $\{0°, 90°, 180°, \text{and } 270°\}$. However, such methods are tailored for image data, and cannot be directly applied to high-dimensional structured data. To address the challenge of rotation on structured data, we propose a self-labeling method through the random rotation [32]. The details are described as follows.

Given a feature space $\mathcal{Z} \subseteq \mathbb{R}^d$, a matrix $R \in \mathbb{R}^{d\times d}$ is called a *proper rotation matrix* when it is orthogonal and has unit determinant, namely

$$R^\top = R^{-1} \quad \text{and } |R| = 1.$$

The set of all proper rotation matrices form the special orthogonal group, which we denote as $SO(d)$. It is a subgroup of the orthogonal group $O(d)$ that contains additional improper rotations involving reflections (orthogonal matrices with determinant -1). To be more specific, matrices in $SO(d)$ share the same determinant 1, whereas matrices in $O(d)$ may have determinant 1 or -1.

We perform the random rotation by uniformly sampling over all feasible rotations in $SO(d)$. It is worth mentioning that uniformly sampling in $SO(d)$ is not equivalent to rotating a random angle in the spherical coordinates. As it can be proved, randomly rotating one angle does not guarantee a uniform distribution over all rotations when the feature dimension d is larger than 2. Actually, some rotations are more likely to be generated than others. Inspired by [32], we adopt the "true" uniformly random rotations instead of naïve ones. Because sampling naïve random rotations that are not uniformly distributed can lead to some rotations being over-represented and others being under-represented in the resulting distribution, causing bias in the solution. Besides, for categorical variables, rotation is unnecessary and ill-defined. Hence, the random rotations will not be applied to categorical variables.

2) Random Fourier Transform: A well-observed phenomenon is that, when the dimension of the input data d is small, the features generated by the random rotation transformation often overlap across categories. This makes the subsequent RF classifier fail to detect anomaly. To solve this problem, we adopt the random approximation method proposed in [25] to first transform the low-dimensional data into a high-dimensional feature space. Specifically, on the input data, we apply a random transformation $\psi : \mathbb{R}^d \to \mathbb{R}^D$, where $D \geq d$. According to [25], when D is large enough, the evaluation of the kernel $\mathcal{K}(\mathbf{x}, \mathbf{y})$ between $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{y} \in \mathbb{R}^d$ can be approximated by the inner product $\psi(\mathbf{x})^\top\psi(\mathbf{y})$. It is shown that the transformed feature space saves the geometric distribution of the data in the original low-dimensional feature space. In this article, we define $\psi(\mathbf{x}) := [\psi_1(\mathbf{x}), \dots, \psi_D(\mathbf{x})]^\top$ and each $\psi_i(\cdot)$ is a real-valued transformation function (see [25])

$$\psi_i(\mathbf{x}) = \sqrt{\frac{2}{D}}\cos\left(\mathbf{w}_i^\top\mathbf{x} + \phi_i\right), \quad i = 1, 2, \dots, D. \tag{1}$$

Here, ϕ_i is uniformly sampled from the interval $[0, 2\pi]$, and $\mathbf{w}_i$ is randomly sampled from the distribution

$p(\mathbf{w}) = \sqrt{2\gamma}\mathcal{N}(\mathbf{w}; 0, 1)$. $\mathcal{N}(\mathbf{w}; 0, 1)$ denotes a standard Gaussian distribution, and γ is the parameter of the approximated RBF kernel. One can easily find that $p(\mathbf{w})$ is the Fourier transform of the Gaussian kernel function. When the dimension of the input data is high enough, this transformation is unnecessary. The parameter γ defines how far the influence of a single training example reaches, with low values meaning "far" (i.e., more disperse the transformed data distribution) and high values meaning "close" (i.e., more compact the transformed data distribution).

3) Theoretical Analysis: This part devotes to a theoretical analysis over why the proposed method can outperform GOAD [24]. More specifically, it proves that the distribution of the self-labeled data obtained by combining RFT together with random orthogonal rotation is more stable than that obtained by a RAT. This stability is measured by the variance of the kernel function within the transformed feature space.

Let $OS(D)$ be the orthogonal group on the transformed feature space of $\mathcal{Z} := \{\psi(\mathbf{x}_1), \ldots, \psi(\mathbf{x}_N)\} \subseteq \mathbb{R}^D$. For any rotation transformation $A \in OS(D)$, we denote the feature after rotation as $r_A(\mathbf{x}) = A\psi(\mathbf{x})$, where $\mathbf{x} \in \mathbb{R}^d$ is the input data. We denote the inner product of $\mathbf{x}$ and $\mathbf{y}$ in the transformed space $\mathbb{R}^D$ with the orthogonal transforms A and B as

$$\mathcal{K}^{\mathrm{AB}}(\mathbf{x}, \mathbf{y}) \doteq r_A(\mathbf{x})^\top r_B(\mathbf{y}). \tag{2}$$

Denote the approximate kernel with ψ taking the RAT as $\mathcal{K}_{\mathrm{RAT}}(\mathbf{x}, \mathbf{y})$ and the approximate kernel with ψ taking the RFT as $\mathcal{K}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})$.

Theorem 1: For any $A, B \in OS(D)$, $\mathcal{K}^{\mathrm{AB}}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})$ is an unbiased estimator of the Gaussian kernel, i.e.,

$$\mathbb{E}\big(\mathcal{K}^{\mathrm{AB}}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})\big) = \frac{\mathrm{tr}(A^\top B)}{D} e^{-\|\mathbf{x}-\mathbf{y}\|^2/(2\sigma^2)}. \tag{3}$$

Let $v = \|\mathbf{x} - \mathbf{y}\|/\sigma$. The variance of $\mathcal{K}^{\mathrm{AB}}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})$ is

$$\mathrm{Var}\big(\mathcal{K}^{\mathrm{AB}}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})\big) = \frac{\mathrm{tr}(A^\top B)}{2 - D}\Big(1 - e^{-v^2}\Big)^2. \tag{4}$$

Proof: From (2), we have

$$\begin{aligned}\mathcal{K}^{\mathrm{AB}}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})) &= r_A(\mathbf{x})^\top r_B(\mathbf{y}) \\ &= \psi(\mathbf{x})^\top A^\top B \psi(\mathbf{y}) = r_I(\mathbf{x})^\top r_U(\mathbf{y})\end{aligned} \tag{5}$$

where $U = A^\top B$ is also a member of the orthogonal group $OS(D)$. Let $\mathbf{u}_i = [u_{i1}, \ldots, u_{iD}]$ be the ith row of U. Then

$$r_I(\mathbf{x})^\top r_U(\mathbf{y}) = \frac{2}{D}\sum_{i,j} u_{ij}\cos\big(\mathbf{w}_i^\top \mathbf{x} + \phi_i\big)\cos\big(\mathbf{w}_j^\top \mathbf{y} + \phi_j\big). \tag{6}$$

The expectation of the approximate kernel is

$$\begin{aligned}&\mathbb{E}[r_I(\mathbf{x})^\top r_U(\mathbf{y})] \\ &\quad = \frac{2}{D}\sum_{i,j} u_{ij}\mathbb{E}\big[\cos\big(\mathbf{w}_i^\top \mathbf{x} + \phi_i\big)\cos\big(\mathbf{w}_j^\top \mathbf{y} + \phi_j\big)\big].\end{aligned} \tag{7}$$

Since $\phi_i \sim \mathrm{Uniform}[0, \phi]$, $\mathbf{w}_i$ is drawn from the Fourier transform $p(\mathbf{w})$, then

$$\begin{aligned}&2\sum_{j=1}^{D} u_{ij}\mathbb{E}_{\mathbf{w}_i,\phi_i,\mathbf{w}_j,\phi_j}\big[\cos\big(\mathbf{w}_i^\top \mathbf{x} + \phi_i\big)\cos\big(\mathbf{w}_j^\top \mathbf{y} + \phi_j\big)\big] \\ &\quad = \sum_{j=1}^{D} u_{ij}\big(\mathbb{E}_{\mathbf{w}_i,\phi_i,\mathbf{w}_j,\phi_j}\big[\cos\big(\mathbf{w}_i^\top \mathbf{x} - \mathbf{w}_j^\top \mathbf{y} + \phi_i - \phi_j\big)\big] \\ &\qquad + \mathbb{E}_{\mathbf{w}_i,\phi_i,\mathbf{w}_j,\phi_j}\big[\cos\big(\mathbf{w}_i^\top \mathbf{x} + \mathbf{w}_j^\top \mathbf{y} + \phi_i + \phi_j\big)\big]\big) \\ &\quad = u_{ii}\mathbb{E}_{\mathbf{w}_i}\big[\cos\big(\mathbf{w}_i^\top(\mathbf{x} - \mathbf{y})\big)\big].\end{aligned} \tag{8}$$

Then, let $\mathbf{v} = (\mathbf{x} - \mathbf{y})/\sigma$, we have

$$\mathbb{E}(\mathcal{K}^{\mathrm{AB}}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})) = \frac{1}{D}\sum_{i=1}^{D} u_{ii}\mathbb{E}\big[\cos(\mathbf{w}_i^\top \mathbf{v})\big]. \tag{9}$$

Let $\mathbf{w}$ be a d-dimensional vector distributed $N(0, I_d)$. By Bochner's theorem,

$$\mathbb{E}[\cos(\mathbf{w}^\top \mathbf{v})] = e^{-v^2/2} \tag{10}$$

and $(1/D)\sum_{i=1}^{D} u_{ii} = \mathrm{tr}(A^\top B)/D$, hence

$$\mathbb{E}(\mathcal{K}^{\mathrm{AB}}_{\mathrm{RFT}}(\mathbf{x}, \mathbf{y})) = \tfrac{1}{D}\textstyle\sum_{i=1}^{D} u_{ii} e^{-v^2/2} = \tfrac{\mathrm{tr}(A^\top B)}{D} e^{-v^2/2} \tag{11}$$

where $v = \|\mathbf{v}\|$. RFT yields an unbiased estimate.

We now compute the variance of RFT approximation. Observe that

$$\cos^2(\mathbf{w}^\top \mathbf{v}) = \frac{1 + \cos(2\mathbf{w}^\top \mathbf{v})}{2} = \frac{1 + \cos(\mathbf{w}^\top(2\mathbf{v}))}{2}. \tag{12}$$

Hence by Bochner's theorem

$$\mathbb{E}[\cos^2(\mathbf{w}^\top \mathbf{v})] = \frac{1 + e^{-2v^2}}{2}. \tag{13}$$

Therefore,

$$\begin{aligned}\mathrm{Var}(\cos(\mathbf{w}^\top \mathbf{v})) &= \mathbb{E}[\cos^2(\mathbf{w}^\top \mathbf{v})] - (\mathbb{E}[\cos(\mathbf{w}^\top \mathbf{v})])^2 \\ &= \frac{1 + e^{-2v^2}}{2} - e^{-v^2} = \frac{(1 - e^{-v^2})^2}{2}.\end{aligned} \tag{14}$$

If we take D such independent random variables $\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_D$, since variance of the sum is sum of variances

$$\mathrm{Var}\left(\frac{1}{D}\sum_{i=1}^{D} u_{ij}\cos(\mathbf{w}_i^\top \mathbf{v})\right) = \frac{\mathrm{tr}(A^\top B)(1 - e^{-v^2})^2}{2 - D}. \tag{15}$$

Theorem 2: Consider the sample data $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^{N}$, where $\mathbf{x}_i \in \mathbb{R}^d$. Suppose that $\mathcal{W}^{D\times d} \subset \mathbb{R}^{D\times d}$ be the affine space of d-to-D-dimensional transformations from the standard normal distribution. For any $A, B \in \mathcal{W}^{D\times d}$ and any $\mathbf{x}, \mathbf{y} \in \mathcal{D}$, let $\mathcal{K}^{A,B}_{\mathrm{RAT}}(\mathbf{x}, \mathbf{y}) \doteq r_A(\mathbf{x})^\top r_B(\mathbf{y}) = (A\mathbf{x})^\top B\mathbf{y}$, then the variance of

$$\mathrm{Var}\Big(\mathcal{K}^{A,B}_{\mathrm{RAT}}(\mathbf{x}, \mathbf{y})\Big) = D\|\mathbf{x}\|^2\|\mathbf{y}\|^2. \tag{16}$$

Proof: Let $C = A^\top B$. Then

$$c_{ij} = \sum_{t=1}^{D} a_{ti} b_{tj}. \tag{17}$$

Since a_{ti} and b_{tj} are independent of each other, the expectation of the product $a_{ti}b_{tj}$ is equal to 0, so we have

$$\mathbb{E}[\mathbf{x}^\top A^\top B\mathbf{y}] = \mathbb{E}\left[\sum_{i,j=1}^{d} x_i c_{ij} y_j\right] = 0. \tag{18}$$

For the expectation of squared quantity, we break it into a summation of terms

$$\mathbb{E}[(\mathbf{x}^\top A^\top B\mathbf{y})^2] = \mathbb{E}\left[\left(\sum_{i,j=1}^{d} x_i c_{ij} y_j\right)^2\right] = \sum_{i,j,k,l=1}^{d} x_i y_j x_k y_l \mathbb{E}[c_{ij}c_{kl}]. \tag{19}$$

It suffices to evaluate $\mathbb{E}[c_{ij}c_{kl}]$ in the following cases.

Case 1: When $i = k$, $j = l$, we have

$$\mathbb{E}[c_{ij}^2] = \mathbb{E}\left[\left(\sum_{t=1}^{D} a_{ti}b_{tj}\right)^2\right] = \sum_{t=1}^{D}\mathbb{E}\left[a_{ti}^2 b_{tj}^2\right] + \sum_{t\neq s}^{D}\mathbb{E}[a_{ti}a_{si}b_{tj}b_{sj}] = D.$$

Case 2: When $i = k$, $j \neq l$, we have

$$\mathbb{E}[c_{ij}c_{il}] = \sum_{t=1}^{D}\mathbb{E}\left[a_{ti}^2 b_{tj}b_{tl}\right] + \sum_{t\neq s}^{D}\mathbb{E}[a_{ti}a_{si}b_{tj}b_{sl}] = 0. \tag{20}$$

Case 3: When $i \neq k$, $j = l$, similarly, $\mathbb{E}[c_{ij}c_{kl}] = 0$.

Case 4: When $i \neq k$, $j \neq l$, similarly, $\mathbb{E}[c_{ij}c_{kl}] = 0$.

In summary, we have

$$\mathrm{Var}\left(\mathcal{K}_{\mathrm{RAT}}^{A,B}(\mathbf{x},\mathbf{y})\right) = D\sum_{i,j=1}^{d} {x_i}^2 {y_j}^2 = D\|\mathbf{x}\|^2\|\mathbf{y}\|^2. \tag{21}$$

From (16) in the Theorem 2, the approximate kernel with RAT $\mathcal{K}_{\mathrm{RAT}}(\mathbf{x},\mathbf{y})$ is linearly positively related to the norm of the input samples $\mathbf{x}$, $\mathbf{y}$ and the dimension D of the transformed space. Comparing (16) and (4), we can observe that when the transformed dimension D increases, the variance of the approximate kernel of the RAT $\mathrm{Var}(\mathcal{K}_{\mathrm{RAT}}^{A,B}(\mathbf{x},\mathbf{y}))$ increases, but the variance of the approximate kernel of the RFT $\mathrm{Var}(\mathcal{K}_{\mathrm{RFT}}^{A,B}(\mathbf{x},\mathbf{y}))$ becomes smaller, which indicates that the data distribution using the RFT is more stable than the RAT.

Our anomaly detection algorithm is built on self-labeled classification data obtained through a set of random transformation mappings and then a classifier is trained to predict the class distribution and construct the sample anomaly scores. Therefore, controlling and generating appropriate data distribution will be beneficial to our self-labeled classification-based anomaly detection. Kernel method is often used to implement nonlinear learning; kernel approximation is a powerful technique to make kernel methods scalable by mapping input features into a new space where inner products approximate the kernel well. Random Fourier features is a typical data-independent technique to approximate the kernel function using an explicit feature mapping RFT [33]. We adopt RFT combined with a set of random orthogonal rotations as random transformation mappings to transform the original data into a new feature space. Theorem 1 shows that the sample inner product in the transformed space can approximate the Gaussian kernel function, and the inner product's expectation variance $\mathrm{Var}(\mathcal{K}_{\mathrm{RFT}}^{\mathrm{AB}}(\mathbf{x},\mathbf{y}))$ is fixed under a given transformation matrix and dimension, which meets the general properties of the Gaussian kernel function. In comparison, Theorem 2 shows that using a set of RAT does not meet the requirements of the kernel function (the expectation of the transformed sample inner product is 0, and the variance of the transformed sample inner product is related to the sample size). In addition, adjusting the parameter γ of RFT is equivalent to adjusting the distribution of the approximate Gaussian kernel, which affects the fitting and generalization ability of the classifier.

A stable data distribution offers two primary advantages for anomaly detection.

1) *Consistency and predictability:* Stable data distributions ensure a lower expected variance in kernel approximations after applying random transformations. This translates to more predictable and less random data distributions, making it easier to discern patterns and detect anomalies. Furthermore, the smooth nature of the kernel approximation, such as the approximated RBF kernel, resists minor variations, enhancing the model's robustness.
2) *Robust feature representation:* Stability in transformations, as offered by RFT, means that features remain consistent, allowing models to focus on inherent data structures rather than noise. This robust representation aids in emphasizing crucial patterns pivotal for anomaly detection, as deviations from these patterns are typical indicators of anomalies.

In essence, a more stable data distribution facilitates better model training, reduces the risk of overfitting, and strengthens the detection of genuine anomalies by prioritizing consistent, noise-resistant features.

4) How to Guarantee the Appropriate Transformation of the Data Distribution: Our method transforms the data distribution by sequentially applying an RFT and then ROTs. The key to this transformation lies in the parameter γ used in the approximated RBF kernel $\exp(-\gamma\|x-y\|^2)$ during the RFT stage. The appropriate data distribution can be achieved by tuning the parameter γ (in our experiments, we tune γ among the values of $\{0.1, 0.01, 0.001\}$). Specifically, γ influences the transformation of the data distribution in the following two ways.

1) *Width of the kernel:* By adjusting γ, we can control the width of the approximated RBF kernel, thereby finding the optimal balance between the model complexity and its generalization ability.
2) *Distribution of* $\mathbf{w}_i$*:* For an RFT, each $\mathbf{w}_i$ in (1) is sampled from a Gaussian distribution $p(\mathbf{w}) = \sqrt{2\gamma}\mathcal{N}(\mathbf{w}; 0, 1)$. Thus, the spread of this Gaussian distribution is influenced by γ. Larger γ leads to wider distributions of $\mathbf{w}_i$, while smaller γ results in more concentrated distributions of $\mathbf{w}_i$.

B. sForest Classifier

In this section, we provide a detailed description on the proposed self-supervised classifier. On the normal training dataset $\mathcal{D} := \{\mathbf{x}_1, \ldots, \mathbf{x}_N\} \subseteq \mathbb{R}^d$, we first conduct feature transformation with the RFT $\psi(\cdot)$. Then, we generate M rotation matrices $\{R_m\}_{m=1}^M$ by uniformly sampling among SO(d) and construct the self-labeled normal dataset as

$$\mathcal{D}^A := \{(R_m(\psi(\mathbf{x}_i)), R_m)\}_{i\in\{1,\ldots,N\}, m\in\{0,\ldots,M\}}. \tag{22}$$

Here, R_0 denotes the identical transformation and $\{R_0(\psi(\mathbf{x}_i))\}_{i\in\{1,\ldots,N\}}$ stand for the data only after the RFT $\psi(\cdot)$. It is worth noting that when the data dimension d is very small (e.g., $d < 5$), it is necessary to do the RFT to transform the data into a higher-dimensional feature space where the dimension of transformed data $D \gg d$ (e.g., $D = 100$). However, when the data dimension d is large or the data contains lots of redundant features, we need to reduce the data dimension before applying the RFT. Reducing the dimensionality of the high-dimensional data can eliminate redundant information contained in the data, thus improving the final performance of the classifier. Moreover, it can efficiently reduce the computational complexity for the subsequent RF training. Here, various dimensionality reduction methods apply, including principle components analysis (PCA), singular-value decomposition (SVD), etc.

Now we are ready to construct our sForest based on the RF classifier proposed by [34]. First, we generate B buckets of self-labeled normal samples randomly through a bootstrap method, with the bth sample bucket denoted as $\mathcal{D}_b \subset \mathcal{D}^A$. Then, we construct a decision tree on each bucket. It divides the predictor space into non-overlapping regions $\{A_j\}_{j=1}^J$, with J denoting the total number of terminal nodes of the decision tree. Borrowing notations from [35], in a node j, representing a region A_j with N_j observations, for $(R(\psi(\mathbf{x}_i)), Y_i) \in \mathcal{D}_b$, let

$$\hat{p}_{j,m,b} := \frac{1}{N_j} \sum_{(R(\psi(\mathbf{x}_i)), Y_i)\in A_j} \mathbf{1}(Y_i = R_m)$$

be the proportion of class R_m observations in node A_j, where the random parameter R stands for random rotation and $\mathbf{1}(\cdot)$ is the indicator function. Then, we take the vector

$$\hat{R}_b(\mathbf{x}; R, \mathcal{D}_b) := \left(\sum_{j=1}^J \hat{p}_{j,m,b} \cdot \mathbf{1}(\mathbf{x} \in A_j)\right)_{m=0}^M$$

as the output of the bth self-supervised tree. We mention that $\hat{R}_b(\mathbf{x}; R, \mathcal{D}_b)$ is a standard vector since $\sum_{m=0}^M \hat{p}_{j,m,b} = 1$.

Then by ensembling the outputs of all trees, the self-supervised forest classifier can be denoted as

$$\hat{R}_{\text{sForest}}(\mathbf{x}; R, \mathcal{D}) := \frac{1}{B} \sum_{b=1}^B \hat{R}_b(\mathbf{x}; R, \mathcal{D}_b). \tag{23}$$

Intuitively, we can take the mth element of vector $\hat{R}_{\text{sForest}}(\mathbf{x}; R, \mathcal{D})$ as the probability of $R(\mathbf{x})$ being classified to R_m. The detailed construction procedure of sForest is presented in Algorithm 1.

Here, we provide an intuitive explanation on why our model can boost the performance of anomaly detection with two toy datasets "blob" and "moons." Both of the two datasets contain 200 data points, 80% of which are normal data and 20% abnormal points generated by uniform sampling in the region of $[-7, 7] \times [-7, 7]$. The normal data points of "blob" are generated by isotropic Gaussian. The normal data of "moons" contains two interleaving half-circles. The two toy datasets are visualized in Fig. 2(a1) and (b1).

On the two datasets, we first increase the data dimension from 2 to 10 with RFT, and then generate different random orthogonal matrices to transform the 10-D features into different categories. We visualize the generated self-labeled data in Fig. 2(a2) and (b2). Different colors in Fig. 2 represent different categories. Note that the abnormal data is specially marked by adding a black edge. Surprisingly, most of the abnormal data is clustered to the center of Fig. 2(a2) and (b2), while the normal data is more concentrated to different clusters according to their corresponding labels. Then the classifier trained by the normal data and their corresponding rotations makes it easier to classify the normal data, and harder for the abnormal ones. Note that the normal clusters also contain a small number of abnormal data, which mainly results from a

Algorithm 1 Construction of sForest

Input: Training (normal) data $\mathcal{D} := \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$;
The random rotation matrices $\{R_m\}_{m=0}^M$ ($R_m \in SO(D)$, R_0 denotes the identical transformation).
Apply the RFT $\psi(\cdot)$ to $\mathcal{D}$ and obtain the data $\{(\psi(\mathbf{x}_0), \ldots, \psi(\mathbf{x}_N))\}$;
for $m = 0 \to M$ **do**
    Generate random rotation matrix R_m;
    Apply R_m to $\{(\psi(\mathbf{x}_0), \ldots, \psi(\mathbf{x}_N))\}$ and achieve labeled data $\{(R_m(\psi(\mathbf{x}_i)), R_m)\}_{i=1}^N$;
end for
Obtain self-labeled normal dataset $\mathcal{D}^A$ as in Eq. (22).
Build random forest classifier based on $\mathcal{D}^A$.
Output: $\hat{R}_{\text{sForest}}(\mathbf{x}; R, \mathcal{D})$ defined as in Eq. (23).

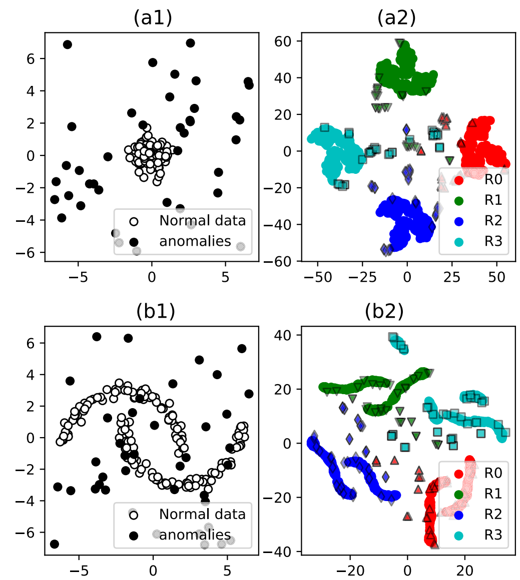

Fig. 2. t-SNE visualization of RFT-based rotated data. (a1) Toy data. (a2) Self-labeled data. (b1) Toy data. (b2) Self-labeled data.

small blob of the generated abnormal samples overlaps with the normal data.

C. Self-Supervised Forest (sForest) for Anomaly Detection

After training the classifier, the main issue is how to separate the anomalous and normal instances in the testing data. Golan and El-Yaniv [17] and Hendrycks and Gimpel [23] claimed that anomalous points tend to obtain lower probability belonging to their predicted labels by a classifier, with the intuition being that class-exclusive features must be captured by training the classifier to distinguish between self-labeled data. Now we apply the well-trained sForest classifier defined in (23) to detect the anomalies.

Let $\mathcal{D}^t = \{\mathbf{x}_1^t, \ldots, \mathbf{x}_N^t\}$ denote the testing dataset that contains both normal samples and anomalies. With RFT, we can obtain the transformed dataset $\{\psi(\mathbf{x}_1^t), \ldots, \psi(\mathbf{x}_N^t)\}$. After the transformation, we apply the rotations $\{R_m\}_{m=0}^M$ to the transformed dataset and generate the self-labeled testing dataset

$$\mathcal{D}^{A,t} := \left\{\left(R_m\left(\psi(\mathbf{x}_i^t)\right), R_m\right)\right\}_{i\in\{1,\ldots,N\}, m\in\{0,\ldots,M\}}. \tag{24}$$

Then, we apply the well-trained sForest classifier to $\mathcal{D}^{A,t}$, and achieve the output vectors

$$\left\{\hat{R}_{\text{sForest}}\left(\psi\left(\mathbf{x}_i^t\right); R_m, \mathcal{D}^t\right)\right\}_{m\in\{0,\ldots,M\}}, \quad i = 1, \ldots, N.$$

Specifically, for each $\mathbf{x}_i^t$, we feed the $M+1$ rotational versions to the classifier, and can simply write the output vectors as an $(M+1) \times (M+1)$ matrix $\hat{A}$

$$\hat{A}(\mathbf{x}_i^t) := \left(\hat{R}_{\text{sForest}}\left(\psi\left(\mathbf{x}_i^t\right); R_m, \mathcal{D}^t\right)\right)_{m=0}^M. \tag{25}$$

Notice that the diagonal elements are the probabilities for every self-labeled $R(\psi(\mathbf{x}_i^t))$ being correctly classified. Therefore, by aiming to categorize the points with low accuracy as anomalies, we design the normality score for $\mathbf{x}_i^t$ as the sum of these probabilities, i.e.,

$$\text{Score}\left(\mathbf{x}_i^t\right) := \text{tr}\left(\hat{A}\left(\psi\left(\mathbf{x}_i^t\right)\right)\right). \tag{26}$$

where $\text{tr}(A)$ means the trace of the matrix A. Normality scores describe how normal a testing sample is, and we recognize those with the lowest normality scores as anomalies. The detailed process of our model for anomaly detection is shown in Algorithm 2.

Algorithm 2 sForest for Anomaly Detection

1: **Input:** The sForest $\hat{R}_{\text{sForest}}(\mathbf{x}; R, \mathcal{D})$ trained as in Eq. (23); Testing data $\mathcal{D}^t = \{\mathbf{x}_1^t, \ldots, \mathbf{x}_N^t\}$; Number of random rotations M; the random rotation matrices $\{R_m\}_{m=0}^M$ ($R_m \in SO(D)$, R_0 denotes the identical transformation).
2: Apply the same RFT $\psi(\cdot)$ employed in training data to $\mathcal{D}^t$.
3: **for** $m = 0 \rightarrow M$ **do**
4:     Apply R_m to $\mathcal{D}^t$ and achieve labeled data $\{(R_m(\psi(\mathbf{x}_i^t)), R_m)\}_{i=1}^N$;
5: **end for**
6: Obtain self-labeled testing dataset $\mathcal{D}^{A,t}$ in Eq. (24);
7: Apply the well-trained $\hat{R}_{\text{sForest}}(\mathbf{x}; R, \mathcal{D})$ classifier to $\mathcal{D}^{A,t}$ and achieve the output matrix $\hat{A}(\mathbf{x}_i^t)$ in Eq. (25).
8: **Output:** Measure the normality and anomaly in the test data based on the score defined as in (26).

IV. Numerical Experiments

In order to evaluate the generalization and effectiveness of the proposed sForest, we conduct extensive experiments on synthetic and real-world datasets. We compare the proposed method with the state-of-the-art methods. Besides, we also carry out extensive numerical simulations to analyze the behavior of the model parameters and data transformation.

A. Baselines

The following anomaly detection methods are taken as baselines for comparison.

1) iForest [26] recognizes anomalies as those more likely to be "isolated" in the early stage of tree construction.
2) PIDForest [28] favors partitions of greatly varying sparsity, and detects anomalies based on the PID scores.
3) GOAD [24] is a classification-based anomaly detector with deep neural networks.
4) LOF [36] is a distance-based anomaly detector focus on the locality of samples.
5) k-NN [37], [38] is a prevailing distance-based anomaly detection method, while it has computational cost when data size and dimension is high.
6) OCSVM [39] trains a classifier to perform the separation of normal points and outliers.
7) COPOD [40], parameter-free method which first constructs an empirical copula, and then uses it to predict tail probabilities of each given data point to determine its level of "extremeness."

B. Experiments on Synthetic Data

In this part, we first demonstrate the advantage of applying a self-supervised learning mechanism to anomaly detection through two synthetic datasets. We conduct the experiment by generating two datasets that each of which contains 400 points, where the proportion of abnormal samples is set to 20%. For the first dataset "moon," the normal points are generated by sklearn with noise setting 0.5, and the second dataset is obtained by employing "make_blobs." For both datasets, the abnormal data are uniformly sampled in the region of $[-7, 7] \times [-7, 7]$.

More specifically, we compare the performance of the proposed method with seven state-of-the-art methods introduced in Section IV-A. For fair comparisons, the number of trees is set as 100 for all the tree-based algorithms, and three fully connected layers are employed for GOAD with the hidden layer setting 100. The default node setting is adapting for k-NN and LOF. For sForest, we first change the dimension to 100 by RFT, and then conduct 16 orthogonal rotations to generate self-supervised labels.

Fig. 3 provides the visual results for different methods on the two datasets, where the normal and abnormal data are, respectively, represented by white and black dots. The red

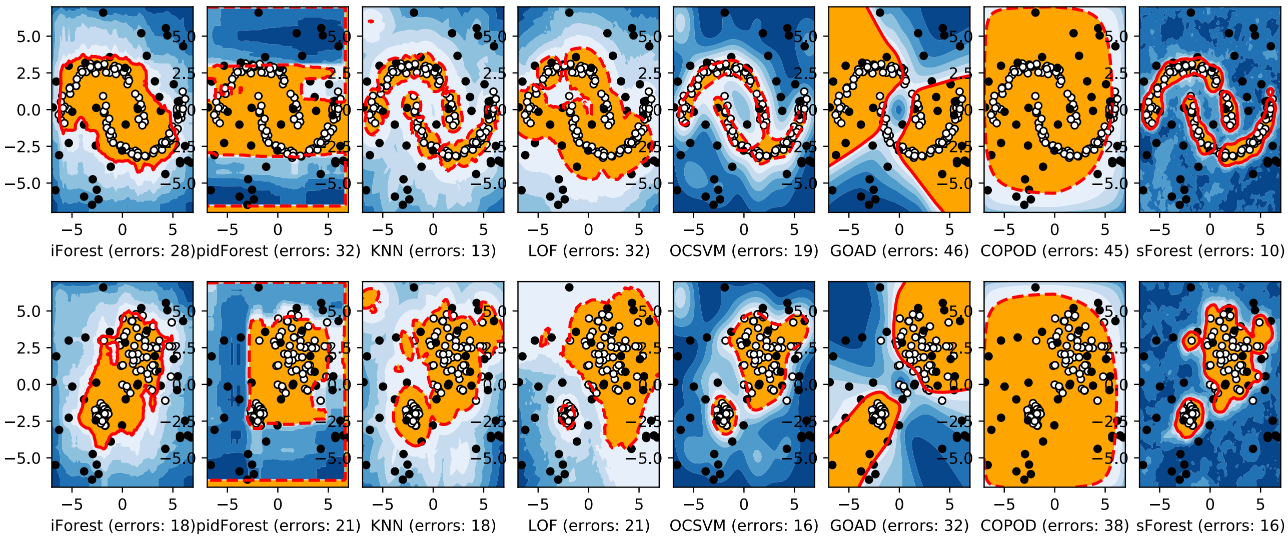

Fig. 3. Comparison of different methods on two synthetic datasets, where the red line denotes the decision boundary for normal and abnormal data.

dotted line is the decision boundary with the yellow and blue regions denoting normal and abnormal data, respectively. Besides, we also provide the degree for abnormal data in the blue region, where the darker region means higher degree of certainty. The number on the downside is the corresponding error for each algorithm for different datasets. Based on the result presented in Fig. 3 that our method could predict the region for normal data more correctly, which is more closed to the ground-truth distribution. In the meanwhile, the lower number of errors can also prove the advantage of our method. Note that our method is consistently and significantly superior to GOAD [24], which is also constructed by the self-supervised mechanism. These synthetic data experiments can also efficiently demonstrate the advantage of our method for anomaly detection compared with other methods.

C. Experiments on Real-World Data

We perform numerical experiments to illustrate the effectiveness of our sForest method compared with other typical anomaly detection algorithms on real-world outlier detection benchmark datasets.

1) Datasets:

1) *Low-dimensional datasets:* We perform numerical experiments on 30 real datasets to illustrate the effectiveness of our proposed sForest on real-world outlier detection benchmark datasets. There are 28 multidimensional real datasets from Outlier Detection DataSets (ODDS), with various sample sizes and dimensionalities. For example, small-size datasets such as `glass`, `ionosphere`, `wbc`, and `wine`, and large-scale datasets such as `forestcover`, `http`, and `smtp` are included in the experiments. High-dimensional datasets such as `arrhythmia` and `speech` are also included. Details of datasets are shown in Table I. For these 28 datasets, our configuration follows that of Bergman and Hoshen [24]. We use 50% of the normal data to train the outlier detector, and evaluate the outlier detector on the remaining 50% of the normal data and all the anomalies.
2) *High-dimensional datasets:* Since the feature dimensions of the above datasets are too low and there are the maximum of 400-D in datasets, we introduce another three datasets with higher dimensions including: 1) `OSQ-768d`; 2) `OSQ-1024d`: these two datasets are generated from a textual dataset `OSQ` in task-oriented dialogue systems. `OSQ` provides a set of manually labeled out-of-scope queries (OSQs) that are not supported by the current system. These OSQs are regarded as anomaly instances in our experiments. We take each utterance as input and obtain its feature vector using the two pre-trained English language models BERT-base and BERT-large [41], respectively; and 3) `IPA` dataset: it contains a set of Chinese utterances that were collected and annotated in the development process of a commercialized intelligent personal assistant (IPA) named Bixby. This dataset covers 81 intents, but 15 intents are not supported by the current IPA system. In this study, these non-supported intents are selected as the out-of-domain intents, and the corresponding utterances are considered as anomaly data. We generate their feature vectors by Chinese BERT-base.[1] The datasets are divided into the train, validation, and test sets, while the anomaly data were only used for validation and test. We keep the same data splits with [42] and [43]. The partitions of the data used in our experiment are shown in Table II.

2) Parameter Settings: In the following, we present the details of implementation. For the proposed sForest, we fix the `number_of_rotations` $M = 64$ and randomly sample each element of $d \times d$ transformation matrices according to the standard normal distribution. For the RF classifier, we fix the `number_of_trees` $B = 100$ and the

[1]https://github.com/google-research/bert

TABLE I
DESCRIPTIONS OF ANOMALY DETECTION BENCHMARK DATASETS

Datasets	#Instances	#Features	#Outliers(%)	Datasets	#instances	#Features	#Outliers(%)
annthyroid	7200	6	534 (7.42%)	musk	3062	166	97 (3.17%)
arrhythmia	452	274	66 (14.60%)	optdigits	5216	64	150 (2.88%)
breastw	683	9	239 (34.99%)	pendigits	6870	16	156 (2.27%)
cardio	1831	21	176 (9.61%)	pima	768	8	268 (34.90%)
forestcover	286048	10	2747 (0.96%)	satellite	6435	36	2036 (31.64%)
glass	214	9	9 (4.21%)	satimage-2	5803	36	71 (1.22%)
heart	267	44	55 (20.60%)	shuttle	49097	9	3511 (7.15%)
http	567498	3	2211 (0.39%)	smtp	95156	3	30 (0.03%)
ionosphere	351	33	126 (35.90%)	speech	3686	400	61 (1.65%)
letter	1600	32	100 (6.25%)	thyroid	3772	6	93 (2.47%)
lympho	148	18	6 (4.05%)	vertebral	240	6	30 (12.50%)
mammo.	11183	6	260 (2.32%)	vowels	1456	12	50 (3.43%)
mnist	7603	100	700 (9.21%)	wbc	378	30	21 (5.56%)
mulcross	262144	4	26214 (10.00%)	wine	129	13	10 (7.75%)

TABLE II
DESCRIPTIONS OF ANOMALY DETECTION BENCHMARK DATASETS. OSQ-768d AND OSQ-1024d HAVE 768- AND 1024-D FEATURES, RESPECTIVELY, WHICH ARE BOTH GENERATED FROM OSQ BY BERT-BASE AND BERT-LARGE. IPA IS CONVERTED INTO 768-D DATA THROUGH BERT-BASE

Datasets	Type	Train	Validate	Test
OSQ	Normal	15000	3000	4500
	Anomaly	250	100	1000
IPA	Normal	28900	3600	3600
	Anomaly	5100	1200	1200

`min_samples_split` $p = 2$. For PIDForest, we use the implementation of the authors provided in github.[2] For iForest, LOF and OCSVM, we utilize the implementation of scikit-learn. For k-NN, we use the PyOD implementation.[3] For hyperparameter settings, we fix the hyperparameters of sub-sampling size to 256 and the number of trees to 100 for iForest. For PIDForest, we fix the hyperparameters of depth to 10 and the number of trees to 50, which is used in [28]. We fix the number of neighbors to 20 for LOF, and use RBF kernel with default hyperparameter gamma for OCSVM, referred to hyperparameter settings in scikit-learn. For k-NN, we fix k to be 5.

The performance evaluation has been done based on the area under the ROC curve (AUC). We run these algorithms 10 times and record the means and standard deviations of the AUC values. Since some methods such as LOF, k-NN, OCSVM, and COPOD are deterministic, it is not necessary to repeat experiments for three datasets (OSQ-768d, OSQ-1024d, and IPA) in which data partitions are provided.

3) Main Results: The experimental results for the low-dimensional datasets are illustrated in Table III. In the aspect of best performance, our method wins 14 out of 31 datasets, and the other three forest-based methods, iForest, PIDforest, win 7 and 5 out of 31 datasets, respectively. The network-based GOAD wins only 1, and the other three deterministic methods, k-NN, LOF, OCSVM, and COPOD, win 6, 6, 2, and 3 out of 31 datasets, respectively. Besides, in the perspective of the top two performance, sForest, iForest, PIDforest, GOAD, k-NN, LOF, OCSVM, and COPOD win 20, 6, 4, 2, 9, 13, 6, and 5 out of 31 datasets. Moreover, in the aspect of the average performance of benchmark datasets, our method has the lowest rank sum.

When it comes to the three high-dimensional datasets as shown in Table IV, sForest consistently outperforms other baselines by a large margin. Specifically, there are 1.3, 4.1, and 8.5 AUC improvements compared to the second best results for OSQ-768d, OSQ-1024d, and IPA, respectively. Note that the performance gap for high-dimensional datasets is more obvious compared to that in low-dimensional datasets. The main reason is that high-dimensional data has more redundant features, and it can boost the performance of the downstream tasks by reducing the dimensionality of the data.

The above results can efficiently demonstrate the advantage of sForest in dealing with tabular data. In the meanwhile, it can also indicate that the self-supervision-based methods can not only achieve the best performance for image data, but also for tabular data. Furthermore, it is also consistent with our analysis that the data distribution obtained by the combination of RFT and orthogonal rotation is more stable and robust. Thus, a better performance is achieved by the proposed sForest.

D. Analysis on Relationship Between Classification and Anomaly Detection

To demonstrate the efficacy of classification-based anomaly detection, we perform simulations on four real datasets of varying sizes and dimensions. We assess the baseline classifiers by calculating the classification accuracy of the self-labeled normal samples in the testing set, each adhering to a specific classification rule. The anomaly detection performance is measured by the area under the ROC curve, denoted as AUC. A higher AUC value signifies greater anomaly detection accuracy.

For parameter settings, we fix the `number_of_rotations` $M = 32$ and the `number_of_trees` $B = 100$ for the all datasets. Then, we vary the `min_samples_split` p to adjust the performance of

[2] https://github.com/vatsalsharan/pidforest
[3] https://github.com/yzhao062/pyod

TABLE III

AUC PERFORMANCE (EXPRESSED IN PERCENTAGES) ON BENCHMARK DATASETS. THE BEST RESULTS ARE MARKED IN **BOLD**, AND THE SECOND BEST IN $\underline{\text{UNDERLINE}}$. FOR sFOREST, iFOREST, PIDFOREST, AND GOAD, THE STANDARD DEVIATION IS REPORTED IN THE SUBSCRIPT. THE LAST ROW SHOWS THE SUMMATION OF RANKS FOR EACH METHOD, WHICH IS THE LOWER THE BETTER. iFOREST [26], PIDFOREST [28], GOAD [24], k-NN [37], LOF [36], OCSVM [39], AND COPOD [40]

Datasets	iForest	PIDforest	GOAD	k-NN	LOF	OCSVM	COPOD	sForest (Ours)
annthyroid	$\underline{91.11}$ $_{1.01}$	**94.47** $_{1.29}$	81.20 $_{5.46}$	71.53 $_{0.31}$	72.16 $_{2.21}$	55.51 $_{1.27}$	77.70 $_{0.39}$	72.13 $_{0.70}$
arrhythmia	79.89 $_{0.90}$	70.78 $_{0.87}$	75.67 $_{2.25}$	$\underline{81.83}$ $_{2.81}$	81.37 $_{1.02}$	57.80 $_{0.65}$	80.90 $_{1.27}$	**82.64** $_{0.46}$
breastw	**99.73** $_{0.03}$	54.04 $_{3.59}$	98.94 $_{0.50}$	$\underline{99.64}$ $_{0.56}$	97.79 $_{6.75}$	99.61 $_{0.53}$	99.39 $_{0.25}$	97.21 $_{1.95}$
cardio	93.39 $_{1.34}$	34.73 $_{5.60}$	92.98 $_{0.22}$	90.62 $_{0.31}$	93.74 $_{1.44}$	**96.71** $_{0.24}$	92.19 $_{0.16}$	$\underline{93.79}$ $_{3.01}$
forestcover	71.93 $_{3.39}$	63.63 $_{1.74}$	17.39 $_{6.87}$	87.96 $_{0.11}$	$\underline{91.21}$ $_{0.44}$	50.03 $_{0.03}$	88.42 $_{0.02}$	**97.51** $_{1.20}$
glass	57.31 $_{1.91}$	$\underline{65.14}$ $_{4.91}$	55.60 $_{0.61}$	62.24 $_{3.55}$	57.93 $_{3.04}$	60.95 $_{2.99}$	64.44 $_{2.19}$	**68.07** $_{1.38}$
heart	$\underline{32.01}$ $_{2.76}$	24.66 $_{1.23}$	30.50 $_{3.39}$	21.11 $_{22.65}$	20.05 $_{2.53}$	23.31 $_{3.27}$	20.99 $_{2.05}$	**43.87** $_{1.39}$
http_mat	99.44 $_{0.16}$	96.26 $_{0.18}$	99.37 $_{0.04}$	99.85 $_{0.00}$	91.86 $_{3.43}$	$\underline{99.85}$ $_{0.00}$	99.16 $_{0.01}$	**99.86** $_{0.02}$
ionosphere	83.71 $_{1.46}$	62.26 $_{3.80}$	90.52 $_{1.13}$	92.50 $_{0.63}$	$\underline{95.10}$ $_{1.39}$	83.71 $_{0.32}$	79.82 $_{2.18}$	**97.29** $_{0.45}$
letter	32.13 $_{1.64}$	$\underline{54.22}$ $_{1.85}$	30.91 $_{0.48}$	36.58 $_{0.87}$	44.04 $_{0.59}$	37.53 $_{0.94}$	**56.06** $_{1.37}$	40.78 $_{4.22}$
lympho	86.03 $_{2.48}$	57.03 $_{7.61}$	90.23 $_{6.66}$	80.87 $_{0.77}$	82.86 $_{1.37}$	83.33 $_{1.02}$	**99.34** $_{0.46}$	$\underline{95.54}$ $_{4.32}$
mammography	$\underline{88.14}$ $_{0.59}$	78.39 $_{2.44}$	70.61 $_{6.48}$	87.24 $_{0.13}$	85.99 $_{1.47}$	87.67 $_{0.10}$	**90.45** $_{0.01}$	82.32 $_{1.89}$
mnist	86.53 $_{1.63}$	68.80 $_{3.90}$	90.26 $_{0.64}$	93.48 $_{0.27}$	$\underline{95.11}$ $_{0.51}$	50.00 $_{0.34}$	77.64 $_{0.38}$	**97.69** $_{0.11}$
mulcross	99.92 $_{0.05}$	96.02 $_{1.43}$	99.90 $_{0.36}$	**100.00** $_{0.00}$	**100.00** $_{0.00}$	**100.00** $_{0.00}$	93.23 $_{0.02}$	99.90 $_{0.18}$
musk	94.49 $_{4.36}$	25.33 $_{7.69}$	**100.00** $_{0.00}$	**100.00** $_{0.00}$	**100.00** $_{0.00}$	50.00 $_{0.00}$	94.76 $_{0.21}$	99.92 $_{0.09}$
optdigits	79.93 $_{3.34}$	88.11 $_{4.40}$	65.00 $_{4.42}$	99.45 $_{0.26}$	$\underline{99.63}$ $_{0.19}$	**99.75** $_{0.85}$	67.85 $_{0.47}$	91.64 $_{0.46}$
pendigits	96.57 $_{1.02}$	95.65 $_{0.89}$	89.17 $_{5.15}$	**99.89** $_{0.07}$	99.11 $_{0.08}$	95.06 $_{0.35}$	90.62 $_{0.38}$	$\underline{99.85}$ $_{0.13}$
pima	**73.30** $_{0.82}$	$\underline{72.22}$ $_{0.92}$	68.38 $_{4.06}$	68.07 $_{0.86}$	66.09 $_{1.67}$	62.28 $_{0.31}$	65.61 $_{0.92}$	64.43 $_{1.54}$
satellite	76.85 $_{1.17}$	74.70 $_{1.02}$	69.07 $_{1.03}$	**82.23** $_{0.23}$	80.12 $_{0.19}$	$\underline{80.65}$ $_{0.20}$	63.40 $_{0.17}$	79.47 $_{0.85}$
satimage-2	99.18 $_{0.16}$	80.86 $_{6.01}$	99.04 $_{0.10}$	$\underline{99.74}$ $_{0.02}$	99.41 $_{0.53}$	92.21 $_{0.01}$	97.45 $_{0.03}$	**99.89** $_{0.03}$
shuttle	99.61 $_{0.10}$	96.30 $_{3.03}$	86.16 $_{7.28}$	99.77 $_{0.01}$	$\underline{99.89}$ $_{0.10}$	99.84 $_{0.05}$	99.45 $_{0.02}$	**99.97** $_{0.00}$
smtp_mat	90.94 $_{1.01}$	90.47 $_{1.15}$	$\underline{91.39}$ $_{2.01}$	**92.75** $_{0.88}$	90.64 $_{2.91}$	83.82 $_{0.13}$	91.21 $_{0.02}$	87.53 $_{3.65}$
speech	38.38 $_{1.59}$	37.17 $_{1.35}$	36.70 $_{1.30}$	36.44 $_{0.66}$	37.59 $_{0.62}$	36.73 $_{0.39}$	$\underline{48.86}$ $_{0.49}$	**49.18** $_{0.55}$
thyroid	**98.93** $_{0.16}$	98.40 $_{0.21}$	96.21 $_{1.24}$	96.08 $_{0.16}$	97.29 $_{2.83}$	86.12 $_{0.19}$	93.88 $_{0.45}$	$\underline{98.90}$ $_{1.27}$
vertebral	14.15 $_{1.40}$	17.52 $_{1.23}$	32.85 $_{15.48}$	12.54 $_{3.60}$	**43.17** $_{4.11}$	26.54 $_{2.91}$	35.01 $_{3.02}$	$\underline{36.10}$ $_{2.85}$
vowels	60.39 $_{4.29}$	62.15 $_{3.72}$	67.43 $_{3.87}$	82.57 $_{0.20}$	**85.64** $_{0.45}$	75.57 $_{0.67}$	49.56 $_{0.58}$	$\underline{82.95}$ $_{3.65}$
wbc	96.58 $_{0.47}$	96.49 $_{0.60}$	96.58 $_{0.34}$	96.20 $_{0.75}$	**97.02** $_{1.38}$	$\underline{96.67}$ $_{0.55}$	96.08 $_{0.21}$	91.45 $_{4.21}$
wine	42.45 $_{12.60}$	37.83 $_{7.95}$	44.18 $_{2.76}$	**100.00** $_{0.00}$	**100.00** $_{0.00}$	96.67 $_{1.06}$	87.67 $_{1.33}$	**100.00** $_{0.00}$
Rank Sum	113	152	142	106	$\underline{95}$	133	131	**80**

TABLE IV

AUC PERFORMANCE (EXPRESSED IN PERCENTAGES) ON THREE TEXTUAL DATASETS. GIVEN THE STABILITY OF AUC PERFORMANCE DUE TO THE LARGE SIZE OF THESE DATASETS, STANDARD DEVIATION IS OMITTED

Datasets	iForest	PIDforest	GOAD	k-NN	LOF	OCSVM	COPOD	sForest
OSQ-768d	55.69	57.24	58.85	73.35	$\underline{75.45}$	55.61	68.11	**76.71**
OSQ-1024d	59.17	59.46	53.86	62.19	62.13	59.95	$\underline{63.39}$	**67.50**
IPA	42.93	38.53	57.31	48.06	$\underline{58.06}$	56.45	43.80	**66.54**

the RF classifier, and observe its impact on the accuracy of anomaly detection. We apply PCA to remove the feature correlation in datasets by transforming the dimensionality of `Mnist` and `Ionosphere` to 32. For `IPA` and `OSQ-678d`, their dimensionality is reduced to 128. We run the experiment five times under different random seeds to obtain the average value of AUC and the classification accuracy.

The results for the four datasets are shown in Fig. 4. As the `min_samples_split` p increases, first the self-label classification accuracy and the AUC criterion for anomaly detection show a negative correlation trend. Meanwhile, as the classification accuracy of the RF classifier decreases, the AUC criterion for anomaly detection turns better. This indicates that choosing an appropriate weak classifier or constructing a more difficult self-supervised classification task is useful for anomaly detection. However, the performance of both classifiers will eventually turn worse consistently. This consistency in the performance of the classification and anomaly detection-based methods can be reasonable properties so that a good classifier is needed to capture the features of the self-labeled normal class. From Fig. 4 for `Mnist`, the classification accuracy drops rapidly, while the anomaly detection AUC declines at a slower pace (e.g., when p increases from 2 to 100, the classification accuracy drops by 10%, but the AUC only decreases by 0.5%). The reason may be that for the self-labeled classification data of `Mnist`, the data distributions of different categories overlap each other, so a classifier with stronger classification ability is required to discriminate. In addition, sForest achieves a high AUC when p is large (indicating the classifier's weaker classification ability). This could be because the abnormal samples are derived from one of the handwritten digits, making them distinctly different from the normal samples.

E. Analysis on the Effect of RFT

In this experiment, we compare the performance between the self-labeled data obtained by the RFT + orthogonal rotation transform and the self-labeled data obtained by the

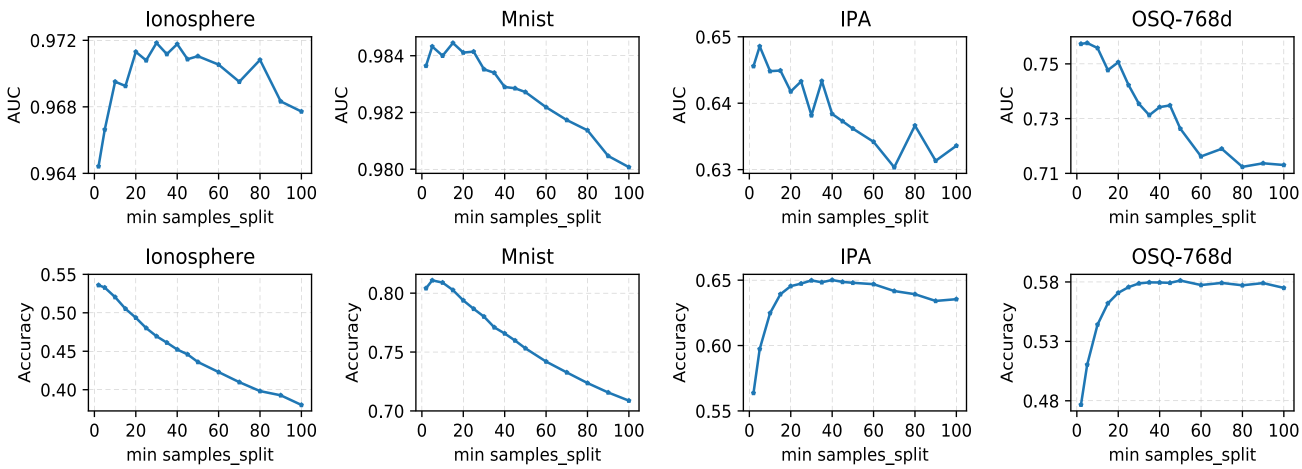

Fig. 4. Relationship between accuracy of classification and anomaly detection.

random affine transform (RAT).[4] For a fair comparison, we set the same setting for both RFT and RAT. We normalize each row of the affine transformation matrix $\mathbf{w}_m$.[5] Note that we denote RFT + orthogonal rotation transform as RFT for simplicity of notation. Specifically, we first generate the training data according to the corresponding setting, where the number of orthogonal rotation transforms is set as 16. Meanwhile, the initial input feature is d, and then is transformed to D. For RFT, we randomly sampled transformation matrices $\mathbf{w}_m$ using the normal distribution for each element. Each matrix has dimensionality $D \times d$. We examine changes in the performance and the trend of the approximation error (MSE) of RBF kernel under different transformation dimensions. The odds from `Ionosphere` and `Mnist` are regarded as test data. In the experiment, the parameter γ of RBF is set to 0.01, and the final experimental results are shown in Fig. 5.

As shown in Fig. 5, the AUC value of sForest is very stable with the increase of transformation dimension when choosing the self-labeled data generated by RFT. This indicates that RFT method can achieve comparable performance without an extra increasing transformation dimension. In addition, the kernel approximation error of this method presents a steady downward trend with the increase of transformation dimension, which indicates that RFT + orthogonal transformation can maintain a good kernel approximation, as well as reflecting the stability of transformation data distribution. However, when it comes to sForest based on RAT, although the kernel approximation error of samples also reduces with the increase of affine transformation dimension, its MSE is far more than the results of RFT. Moreover, the performance fluctuates a lot with different dimensions and has a larger prediction variance. These experiments can indicate the high

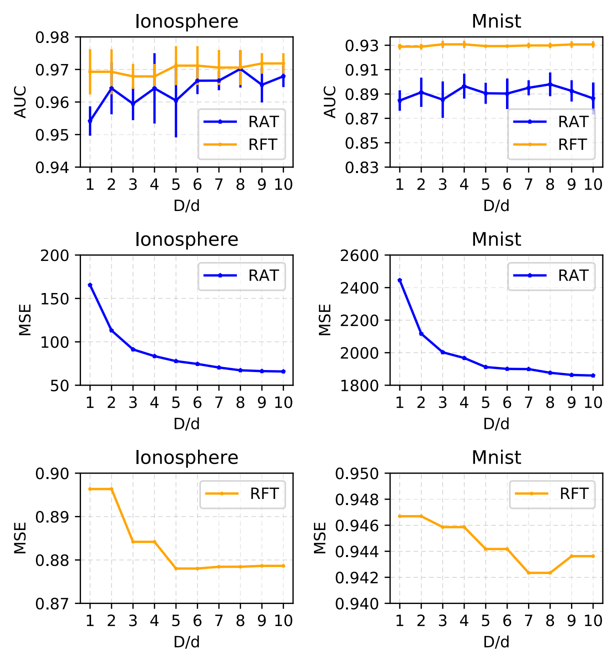

Fig. 5. Comparison of RFT and RAT with the same setting.

effectiveness of generating proper data distribution by our method.

F. Effect of Different Classifiers and Transformations

We conduct the comparison by replacing the RF classifier and our random transformations RFT with other classifiers and RAT, respectively. More specifically, we mainly select Gaussian Naive Bayesian (GNB), logical regression (LR), support vector machine (SVM), k-NN, and multilayer perceptrons (MLPs) as the base classifiers. The final results are shown in Table V, where different combinations are constructed to compare the performance with six datasets including `satellite`, `arrhythmia`, `ionosphere`, `mnist`, `OSQ-768d`, and `IPA`.

The table clearly indicates that both the choice of classifier and the corresponding data distribution significantly influence the final performance. Notably, when combined with various

[4] This self-labeled data is done by transforming data $X \in \mathbb{R}^{n\times d}$ using M different affine transformations into $T(\mathbf{x}, m) = \mathbf{w}_m X^\top$, where $\mathbf{w}_m \in \mathbb{R}^{D\times d}$ is a random matrix.

[5] If normalization is not performed, the MSE of its kernel approximation shows a rapid increase with the increase of D/d. After normalizing $\mathbf{w}_m$, the kernel approximation MSE of RAT shows a decreasing trend with the increase of D/d. Besides, we also find that normalizing or not normalizing $\mathbf{w}_m$ has no significant effect on its anomaly detection results.

TABLE V

EXPERIMENTAL RESULTS OF THE PROPOSED ALGORITHM UNDER DIFFERENT CLASSIFIERS AND TRANSFORMATIONS (RAT AND RFT). THE AUC PERFORMANCE IS EXPRESSED IN PERCENTAGES, AND THE SUBSCRIPT REFER TO THE STANDARD DEVIATION OF THE REPEATED EXPERIMENTS

Dataset	GNB + RAT	GNB + RFT	k-NN + RAT	k-NN + RFT	LR + RAT	LR + RFT
arrhythmia	48.11 $_{2.89}$	**74.56** $_{1.52}$	56.50 $_{0.97}$	**72.97** $_{1.17}$	61.69 $_{1.25}$	**76.81** $_{5.04}$
satellite	74.60 $_{0.75}$	**75.34** $_{0.36}$	50.88 $_{0.09}$	**77.33** $_{0.54}$	63.30 $_{5.35}$	**75.68** $_{3.05}$
ionosphere	75.31 $_{2.87}$	**93.04** $_{1.13}$	53.05 $_{1.38}$	**94.88** $_{0.32}$	59.53 $_{1.30}$	**87.39** $_{5.50}$
mnist	62.50 $_{2.98}$	**84.77** $_{0.64}$	56.56 $_{0.33}$	**91.31** $_{0.15}$	81.61 $_{2.55}$	**89.02** $_{1.92}$
OSQ-768d	46.93 $_{0.28}$	**57.80** $_{0.25}$	50.09 $_{0.09}$	**63.84** $_{0.30}$	49.61 $_{0.87}$	**58.48** $_{4.12}$
IPA	55.16 $_{1.68}$	**59.81** $_{0.21}$	55.18 $_{0.99}$	**74.31** $_{0.46}$	50.47 $_{8.83}$	**59.46** $_{3.64}$
Dataset	MLP + RAT	MLP + RFT	SVM + RAT	SVM + RFT	RF + RAT	RF + RFT
arrhythmia	62.01 $_{1.58}$	**77.08** $_{5.12}$	69.20 $_{6.77}$	**77.22** $_{7.50}$	70.32 $_{0.90}$	**76.40** $_{2.51}$
satellite	61.65 $_{4.24}$	**74.80** $_{2.86}$	**78.24** $_{8.91}$	73.95 $_{1.75}$	**81.39** $_{0.90}$	79.74 $_{0.58}$
ionosphere	59.21 $_{1.39}$	**88.89** $_{5.28}$	66.09 $_{2.97}$	**88.37** $_{4.92}$	90.57 $_{0.78}$	**96.02** $_{0.72}$
mnist	81.44 $_{1.78}$	**89.08** $_{3.03}$	77.57 $_{10.29}$	**90.03** $_{5.03}$	87.28 $_{0.60}$	**90.79** $_{0.51}$
OSQ-768d	60.52 $_{1.32}$	**64.90** $_{2.95}$	53.59 $_{9.06}$	**57.51** $_{1.86}$	64.77 $_{0.45}$	**67.80** $_{1.04}$
IPA	60.85 $_{1.96}$	**61.84** $_{3.15}$	55.56 $_{9.31}$	**56.55** $_{1.55}$	**66.26** $_{1.00}$	63.70 $_{0.94}$

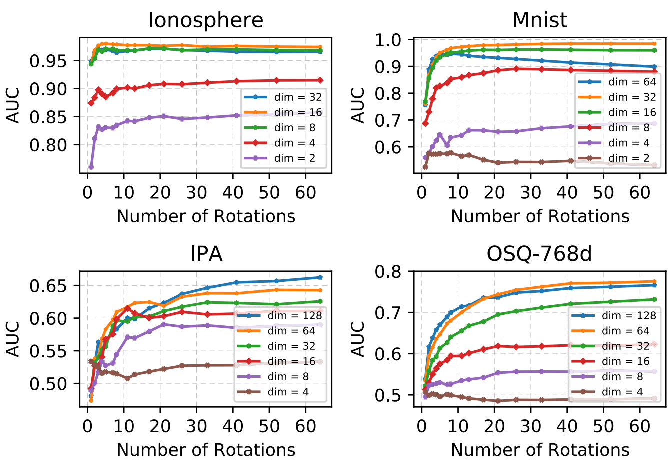

Fig. 6. AUC score versus the `number_of_rotations` M for datasets with different dimensions.

classifiers, RFT consistently outperforms RAT. Specifically, when employing GNB, k-NN, LR, and MLP as classifiers, RFT demonstrates superiority across all datasets in comparison to RAT. Furthermore, RF appears to be more competitive relative to the other five classifiers. It is worth noting that in this section, we only replace the RF and RFT with MLP and RAT, which is inconsistent with GOAD.

G. Analysis on Relationship Between the Dimension of Data and the Number of Rotations

For the purpose of investigating the relationship between the best `number_of_rotations` M and the dimension of dataset, we conduct experiments among datasets with different dimensions. Here, we employ the PCA for dimensionality reduction, and generate datasets with different dimensions among $\{2, 4, 8, 16, 32, 64\}$. The four datasets used in Section IV-D are selected for this experiment. We conduct anomaly detection with our sForest algorithm with different M, and evaluate its performances via AUC. The plots of the AUC score versus the `number_of_rotations` M for datasets with different dimensions are provided in Fig. 6.

As is shown in Fig. 6, we can observe the following conclusions: 1) the performances of anomaly detection would be decreased when the dimensionality of datasets d is too low, such as dim $= 2$ for `Mnist` and `Ionosphere`, dim $=$ 4 for the other two datasets; 2) for the difficult anomaly detection tasks `IPA` and `OSQ-768d`, as the dimension of datasets d becomes higher, the optimal M becomes larger. For example, when the dimension d' of the PCA-transformed `OSQ-768d` dataset is 8, the best M lies in the range [20, 30]. In addition, when the dimension d' raises up to 16 or more, it can consistently boost the performances when M increases, and become steady when M is sufficiently large; 3) for easy anomaly detection datasets `Ionophere` and `Mnist`, the best Ms lie in the range (0, 10], which are relatively small. This indicates that choosing a small M is enough to capture their data information and achieve better AUC; and 4) from dim $=$ 64 in `Mnist`, the AUC value eventually shows a decreasing trend as the `number_of_rotations` M becomes larger. The reason may be that the PCA-transformed features with higher dimensions contain much redundant information. As the number of rotations M increases, the negative impact of this redundant information will be amplified which will further reduce or suppress the accuracy of anomaly detection algorithms. Therefore, before using sForest, it is necessary to perform feature selection or feature learning on the datasets to reduce the redundant information or the correlation of features.

H. Adversarial Samples Experiment

1) Setting: Let us assume an adversarial model in which the attacker knows the model architecture and normal training data with attempting to modify normal samples to become anomalies. We examine three settings: 1) the attacker knows random orthogonal transformation (ROT) and the classifier, but not know the RFT; 2) the attacker knows the classifier, but not RFT and ROT; and 3) the attacker does not know RFT, ROT, and the classifier. Since adversarial samples are mainly targeted at neural network models, we first choose an MLP as the classifier to obtain adversarial samples and test the model's adversarial robustness.

1) To evaluate the benefits of the components of our proposed anomaly detection framework, we construct four self-supervised anomaly detection models using MLP as the classifier: **sMLP1** (RFT + ROT + MLP), **sMLP2** (RFT + ROT + MLP), **sMLP3** (RFT + ROT + MLP), and **sMLP4** (RFT + ROT + MLP). More specifically, sMLP2 has the same ROT and MLP as sMLP1 and sMLP3 has the same MLP as sMLP1. For sMLP4, it has different RFT, ROT, and MLP from sMLP1. We use the same color to indicate the same components used, where the same MLP refers to the same network structure and initialization.
2) We also construct three self-supervised anomaly detection models with RF as the classifier: **sForest1** (RFT + ROT + RF), **sForest2** (RFT + ROT + RF), and **sForest3** (RFT + ROT + RF). In detail, sForest1 has the same RFT and ROT as sMLP1, but with different classifiers; sForest2 has the same ROT as sMLP1 but with different RFTs and classifiers; sForest3 has different RFTs, ROTs, and classifiers from sMLP1.
3) In addition, we also compare the performance with incorporating RAT-based methods including RAT + MLP, RAT + MLP, and RAT + RF. Specifically, RAT + MLP has the same MLP as sMLP1, while RAT + MLP has different MLPs from sMLP1. For RAT + RF, it employs RF as the classifier.

In the experiment, sMLP1 is employed to generate adversarial samples. The number of random orthogonal rotations M is set to 32, and the transformation dimension of the RFT is set to 200. The parameter γ of the RFT for each dataset takes its optimal parameters used in Section IV-C. The MLP is a three-layer perception with hidden size 256 and ReLU activation function. Adversarial examples are created using projected gradient descent algorithm (PGD) [44] based on sMLP1 (making normal samples appear like anomalies). Divide the original test set into two parts: test normal data and test abnormal data, and then combine the adversarial samples obtained from sMLP1 on test normal data with test abnormal data as the final adversarial sample test set to evaluate various models. Here, we run 40 iterations of PGD as our adversary, with a step size of $\alpha = 0.01$ and against perturbations of size $\epsilon = 0.1$. The increase of false classification rate on the adversarial examples is measured using the above models.

Besides, we also design a similar new experiment with the above experiment and test whether our proposed method, using RFT + ROT, has superiority in adversarial robustness compared to RAT. We used another model with RAT transformation, named RAT + MLP, to generate adversarial samples, which were then applied to other model variations including RAT + RF, RFT + ROT + RF, etc. The RATs are obtained through the standard normal distribution generation function numpy.random.randn(). Note that it is similar to the above experimental setting in this experiment, where the same color indicates the same component.

2) Results: The experimental results on six datasets are shown in Tables VI and VII. From the results, we can draw the following conclusions.

1) The AUC values of the models in the first row of both tables significantly decrease under the attack of their generated adversarial samples, indicating that these adversarial samples can effectively attack the model.
2) When the generated adversarial samples are applied to models with significant differences, the adversarial samples basically become ineffective. For example, in Table VI, the adversarial samples generated by RFT + ROT + MLP applied to the model using RAT, and in Table VII, the adversarial samples generated by RAT + MLP applied to the model using RFT + ROT, show that the model performance is largely unaffected, with both increases and decreases on different datasets. This result also aligns with our intuition.
3) In Table VI, the AUC value of RFT + ROT + Forest still shows a significant decrease under adversarial samples. This model shares the same RFT + ROT transformation with RFT + ROT + MLP, but even with a different classifier, RF, the performance decline of other models with different RFTs is very small, mostly ranging between 0% and 2%. This indicates that RFT + ROT can enhance the model's robustness against adversarial samples.
4) In Table VII, the adversarial samples of RAT + MLP applied to other models using RAT still show a significant decrease in performance. The reduction generally exceeds 2%, even if the model uses a different classifier, RF. Compared to Table VI, it is evident that using RFT + ROT provides more advantages in terms of adversarial robustness than using RFT.

I. Contamination Experiment

1) Setting: Our proposed model assumes that the training data are all normal samples. Here, we conduct a contamination experiment to observe the robustness of the model when some anomalies are given normal labels and mixed into the training data. Different from the previous experiment setup, here the anomalies are divided into train and test parts. The train anomalies are all used for data contamination, and the test anomalies are all used for evaluation. We mix the original training set's $1\%, 2\%, \ldots, 10\%$ degree of anomalies into the training data and observe the model's anomaly detection AUC value. Since some datasets have relatively few training anomalies, such as `OSQ-768d`, which is less than 2% (250/15 000), we resample from the training anomalies to obtain the corresponding ratio of contamination samples.

2) Results: Fig. 7 shows the experimental results of the proposed framework under two classifiers RF and MLP, denoted as RFT + Forest (sForest) and RFT + MLP (sMLP). Each experiment is executed five times using different random seeds. From Fig. 7(a) and (b), we can observe that: 1) the AUCs of sForest and sMLP show stable changes on almost all datasets except for `mnist`, which indicates that the proposed framework is generally robust to the degree of contamination and 2) the AUC value of sForest fluctuates steadily and insignificant, but the AUC value of sMLP fluctuates greatly. This indicates that sForest is more stable in anomaly detection than the MLP-based sMLP overall.

TABLE VI

EXPERIMENTAL RESULTS (EXPRESSED IN PERCENTAGES) ON ADVERSARIAL ROBUST OF OUR METHOD. *adv* AND *no_adv* REPRESENT THE EVALUATION RESULTS OF THE MODEL UNDER THE ADVERSARIAL SAMPLE TEST SET AND THE ORIGINAL TEST SET (WITHOUT ADVERSARIAL SAMPLES), RESPECTIVELY. THE VALUE WITH ↓ DENOTES A DECREASE IN AUC UNDER ADVERSARIAL SAMPLES, WHILE ↑ INDICATES AN INCREASE. THE MODEL IN THE FIRST ROW WITH A LIGHT BLUE BACKGROUND IS USED TO GENERATE ADVERSARIAL SAMPLES AND IS APPLIED TO TEST THE REMAINING MODELS. THE BEST RESULTS ARE MARKED IN BOLD

model	arrhythmia		satellite		ionosphere		mnist		OSQ-768d		IPA	
	no_adv	adv	no_adv	adv	no_adv	adv	no_adv	adv	no_adv	adv	no_adv	adv
RFT+ROT+MLP	76.05	4.42↓	77.00	8.78↓	89.17	1.30↓	88.71	4.10↓	65.08	27.62↓	58.21	49.14↓
RFT+ROT+MLP	77.18	0.74↓	76.58	1.98↓	87.83	0.94↓	88.31	0.98↓	64.06	1.46↓	57.96	0.45↓
RFT+ROT+MLP	77.45	0.78↓	74.82	2.28↓	89.05	0.89↓	87.16	1.41↓	64.84	1.36↓	58.57	0.45↓
RFT+ROT+MLP	76.48	0.73↓	77.12	2.17↓	87.83	0.89↓	89.32	1.15↓	63.87	1.33↓	57.11	0.46↓
RFT+ROT+RF	76.63	3.32↓	79.97	3.46↓	**95.77**	0.69↓	**90.51**	2.32↓	**67.37**	9.10↓	60.61	38.29↓
RFT+ROT+RF	77.22	0.73↓	79.18	1.86↓	95.63	0.47↓	89.50	1.37↓	67.22	0.92↓	61.57	0.68↓
RFT+ROT+RF	**77.49**	0.73↓	79.13	1.83↓	95.37	0.56↓	89.84	1.32↓	67.27	0.93↓	61.18	**0.21**↓
RAT+MLP	64.44	**0.48**↑	63.43	0.34↓	55.33	1.49↑	80.49	**1.05**↑	58.76	0.45 ↓	61.26	0.36↓
RAT+MLP	63.50	0.55↑	60.85	0.34↓	55.77	**1.57**↑	81.04	0.66↑	60.53	**0.13**↓	61.19	0.33↓
RAT+RF	69.02	0.00↓	**82.33**	1.22↓	91.89	0.20↑	88.07	0.08↓	64.93	0.19↓	**66.01**	0.25↓

TABLE VII

EXPERIMENTAL RESULTS ON ADVERSARIAL ROBUST OF OUR METHOD. *adv* AND *no_adv* REPRESENT THE EVALUATION RESULTS OF THE MODEL UNDER THE ADVERSARIAL SAMPLE TEST SET AND THE ORIGINAL TEST SET (WITHOUT ADVERSARIAL SAMPLES), RESPECTIVELY. THE VALUE WITH ↓ DENOTES A DECREASE IN AUC UNDER ADVERSARIAL SAMPLES, WHILE ↑ INDICATES AN INCREASE. THE MODEL IN THE FIRST ROW WITH A LIGHT BLUE BACKGROUND IS USED TO GENERATE ADVERSARIAL SAMPLES AND IS APPLIED TO TEST THE REMAINING MODELS. THE BEST RESULTS ARE MARKED IN BOLD

model	arrhythmia		satellite		ionosphere		mnist		OSQ-768d		IPA	
	no_adv	adv	no_adv	adv	no_adv	adv	no_adv	adv	no_adv	adv	no_adv	adv
RAT+MLP	63.98	8.33↓	60.29	4.09↓	55.21	4.14↓	80.28	13.37↓	59.33	35.62↓	60.31	29.81↓
RAT+MLP	64.79	4.99↓	60.95	3.69↓	55.17	4.03↓	79.14	6.13↓	59.15	1.74↓	60.95	2.38↓
RAT+MLP	65.56	4.98↓	61.98	3.53↓	55.47	4.17↓	79.11	6.38↓	59.12	1.62↓	61.12	2.40↓
RAT+RF	68.76	3.15↓	**82.26**	2.17↓	91.99	2.29↓	88.31	2.25↓	64.72	1.15↓	66.10	1.21↓
RAT+RF	68.82	3.20↓	82.10	2.20↓	92.09	2.22↓	87.61	2.30↓	64.73	1.02↓	**66.68**	1.08↓
RFT+ROT+MLP	76.48	0.11↑	76.64	0.61↓	87.01	0.42↑	85.95	0.13↑	64.16	0.20↓	58.06	**0.08**↓
RFT+ROT+MLP	76.55	**0.12**↑	76.72	**0.59**↓	87.16	**0.43**↑	86.55	**0.13**↑	64.16	**0.19**↓	57.79	0.10↓
RFT+ROT+RF	**76.55**	0.05↑	79.97	0.90↓	**95.35**	0.07↑	**89.67**	0.02↑	**66.70**	0.30↓	61.50	0.34↓

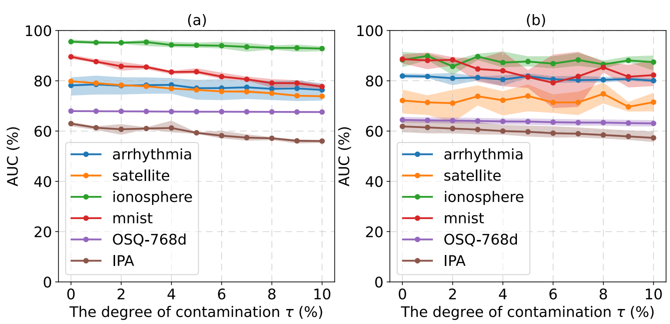

Fig. 7. Plots of the relationship between the degree of contamination and the anomaly detection accuracy (measured by AUC) on six datasets. The proposed sForest is generally robust to the degree of contamination. Each line plot corresponds to the average AUC values, and the shaded region around each line represents the range (from minimum to maximum) of AUC values observed. (a) sForest. (b) sMLP.

V. CONCLUSION

In this article, we solve the classification-based anomaly detection by proposing a more suitable data distribution, which is created for general data with RFT combined with orthogonal rotation transform. Specifically, the training data is first transformed to a new feature space by RFT, and then the feature is constructed to the self-supervised training date by orthogonal rotation transform. Furthermore, a theoretical description is given to show that the proposed method produces more stable data distribution (kernel approximation with lower expected variance) rather than RAT. We conduct different experiments on some artificial and real data to evaluate the performance of the proposed method is more stable and efficient than the distance-based, kernel-based, forest-based, and network-based methods, where the RF classifier is employed to connect the data distribution.

In the future, we aim to extend our method to other domain such as the anomaly detection for time series. Besides, this work is designed for semi-supervised anomaly detection, where the given training data only contain normal samples. In the future work, we plan to design an algorithm for unsupervised anomaly detection, in which both normal and abnormal data are contained in the training data.

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